(7) $5x+2y=3$; Solve using matrix.

$2x+3y+z=1$

$x+4y+3z=7$

→ We know,

$AX=B$

here,

$$A=\begin{bmatrix}5&2&0\\2&3&1\\1&4&3\end{bmatrix},\ X=\begin{bmatrix}x\\y\\z\end{bmatrix},\ B=\begin{bmatrix}3\\1\\7\end{bmatrix}$$

$$\therefore |A|=\begin{vmatrix}5&2&0\\2&3&1\\1&4&3\end{vmatrix}$$

$$=5(9-4)-2(6-1)$$

$$=25-10=15\neq0$$

$$\therefore \text{coeffecient matrix}=\begin{bmatrix}5&-5&5\\-6&15&18\\2&-5&11\end{bmatrix}$$

$$\therefore \text{Adj of }(A)=\begin{bmatrix}5&-5&5\\-6&15&18\\2&-5&11\end{bmatrix}^T$$

$$=\begin{bmatrix}5&-6&2\\-5&15&-5\\5&18&11\end{bmatrix}$$

$$\therefore A^{-1} = \frac{1}{15}\begin{bmatrix} 5 & -6 & 2 \\ -5 & 15 & -5 \\ 5 & 18 & 11 \end{bmatrix}$$

Now,

$$AX = B$$

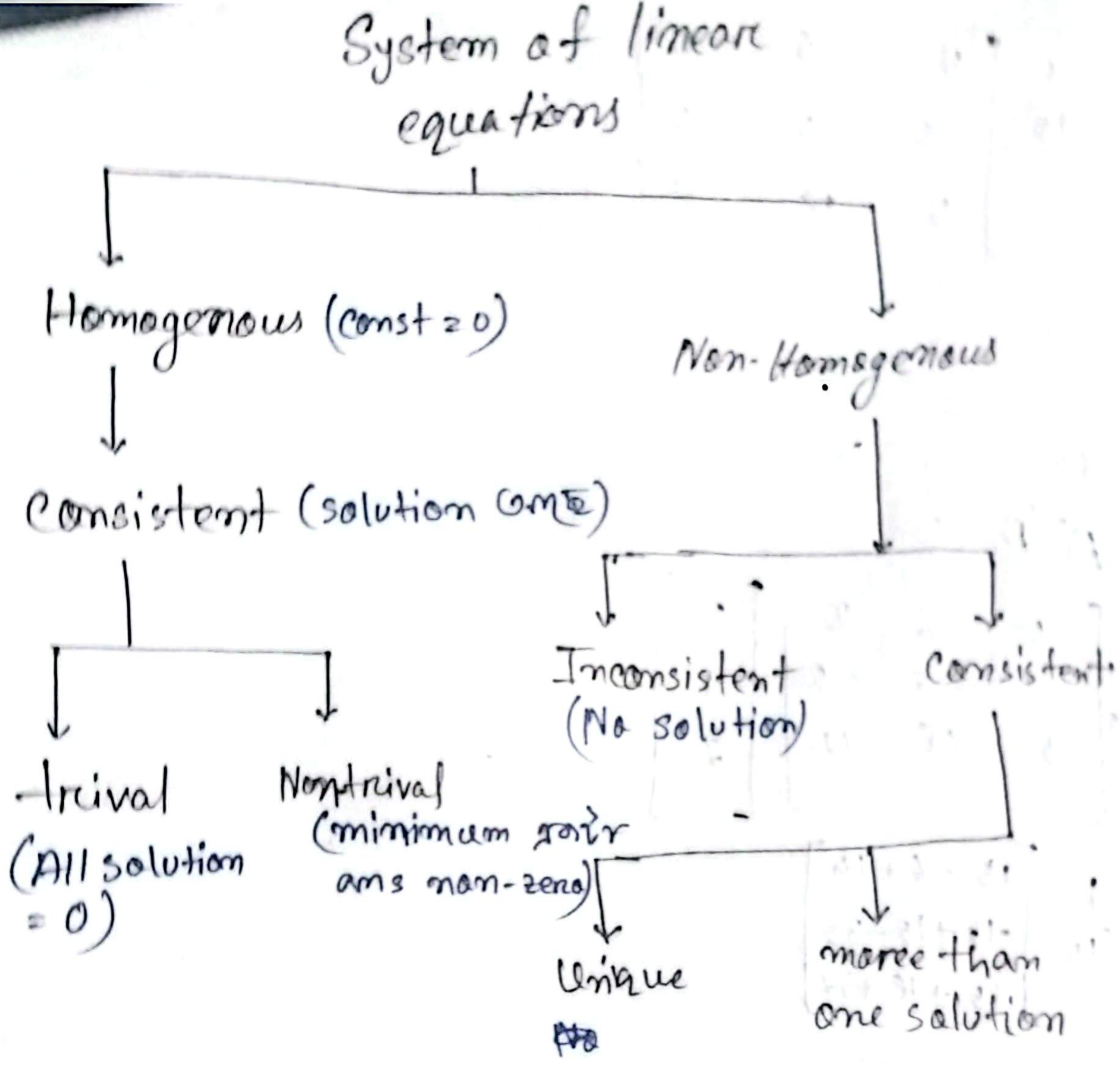
$$\Rightarrow X = A^{-1}B$$

$$= \frac{1}{15}\begin{bmatrix} 5 & -6 & 2 \\ -5 & 15 & -5 \\ 5 & 18 & 11 \end{bmatrix}\begin{bmatrix} 3 \\ 1 \\ 7 \end{bmatrix}$$

$$= \frac{1}{15}\begin{bmatrix} 15-6+14 \\ -15+15-35 \\ 15+18+77 \end{bmatrix}$$

$$= \frac{1}{15}\begin{bmatrix} 23 \\ -35 \\ 110 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \frac{23}{15} \\ -\frac{7}{3} \\ \frac{22}{3} \end{bmatrix}$$


System of linear equations
Homogenous (const = 0)
Non-Homogenous
Consistent (solution আছে)
Inconsistent (No solution)
Consistent
Trival (All solution = 0)
Nontrival (minimum একটির ans non-zero)
Unique
more than one solution

HW:

- Solve the following system of linear equation:
  by using chelon form and matrix form:

$$x + 2y - z = 3$$

$$x + 3y + z = 5$$

$$3x + 8y + 4z = 17$$

$\to e = \frac{}{2}$

$= 30$ Ans

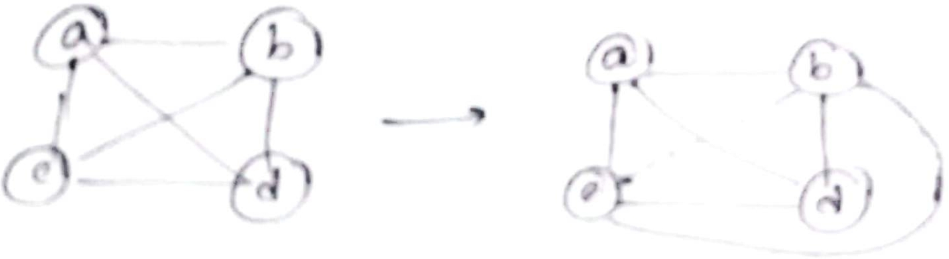
C.W

12.10.2026

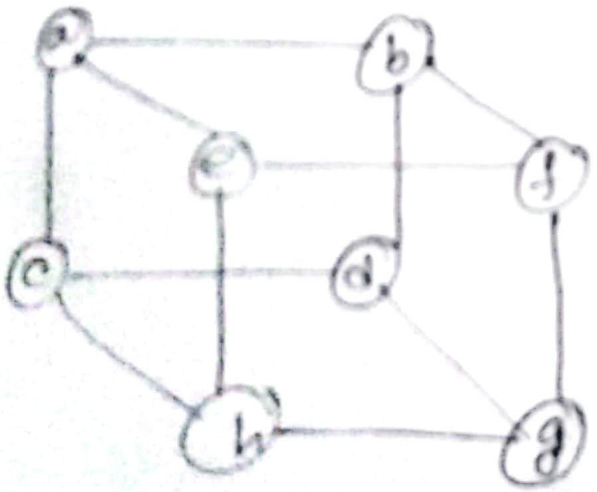
Planner Graph:

- যেখানে কোনো অবস্থায় দুটি edge intercross করবে না।
- x-axis and y-axis - এই - আঁকবে,
- Main fig থেকে আর redraw করবে,

Ex:



Q:



: turn into planner graph.